Class 10

1 \ 2	L	R
T (1/4)	4	1
B (3/4)	2	3

$\left[\frac{1}{4}(T), \frac{3}{4}(B)\right]$

- Suppose G is a strategic form game where all the players have finitely many pure strategies. Then $\forall\, i, j \in N$

$$U_i\left(\lambda \sigma_j + (1-\lambda)\sigma_j', \sigma_{-j}\right) = \lambda U_i\left(\sigma_j, \sigma_{-j}\right) + (1-\lambda) U_i\left(\sigma_j', \sigma_{-j}\right)$$

Proof $U_i\left(\lambda \sigma_j + (1-\lambda)\sigma_j', \sigma_{-j}\right)$

$S = \prod_{i=1}^{n} S_i$

$$= \sum_{(s_1,\ldots,s_n)\in S} u_i(s_1,\ldots,s_n)\, \sigma_1(s_1)\,\sigma_2(s_2)\cdots\left(\lambda\sigma_j(s_j) + (1-\lambda)\sigma_j'(s_j)\right)\cdots\sigma_n(s_n)$$

$$= \sum_{(s_1,\ldots,s_n)\in S} u_i(s_1,\ldots,s_n)\, \sigma_1(s_1)\,\sigma_2(s_2)\cdots\lambda\sigma_j(s_j)\cdots\sigma_n(s_n) + \sum_{(s_1,\ldots,s_n)\in S} u_i(s_1,\ldots,s_n)\, \sigma_1(s_1)\cdots(1-\lambda)\sigma_j'(s_j)\cdots\sigma_n(s_n)$$

$$= \lambda \sum_{(s_1,\dots,s_n)\in S} u_i(s_1,\dots,s_n)\sigma_1(s_1)\cdots\sigma_j(s_j)\cdots\sigma_n(s_n)$$

$$+ (1-\lambda)\sum_{(s_1,\dots,s_n)} u_i(s_1,\dots,s_n)\sigma_1(s_1)\cdots\sigma_j'(s_j)\cdots\sigma_n(s_n)$$

$$= \lambda v_i(\sigma_1,\dots,\sigma_j,\dots,\sigma_n) + (1-\lambda) v_i(\sigma_1,\dots,\sigma_j',\dots,\sigma_n)$$

Corollary $\quad v_i(\sigma_i,\sigma_{-i}) = \sum_{s_i\in S_i} v_i(s_i,\sigma_{-i})\sigma_i(s_i)$

$\sigma_i = \sigma_i(s_1)(1,0,\dots,0) + \sigma_i(s_2)(0,1,\dots,0) + \cdots + \sigma_i(s_{m_i})(0,0,\dots,1)$

1928

Theorem 1 (Von Neumann's Minimax Theorem)

Every two player zero sum game where both players have finitely many pure strategies, have a value in mixed strategies.

(John Nash, 1950)

Theorem Every strategic form game where all the players have finitely many pure strategies have a NE in mixed strategies.

Example

$N = \{1, 2\}$

$S_1 = S_2 = \mathbb{N}$

$$u_1(s_1, s_2) = \begin{cases} 1 & \text{if } s_1 > s_2 \\ 0 & \text{if } s_1 = s_2 \\ -1 & \text{if } s_1 < s_2 \end{cases}$$

$\underline{v} = \max_{s_1 \in \mathbb{N}} \min_{s_2 \in \mathbb{N}} u_1(s_1, s_2) = -1$

$\bar{v} = \min_{s_2 \in \mathbb{N}} \max_{s_1 \in \mathbb{N}} u_1(s_1, s_2) = 1$

$\underline{v}^M = \max_{\sigma_1 \in \Sigma_1} \min_{\sigma_2 \in \Sigma_2} u_1(\sigma_1, \sigma_2)$

$\Sigma_1 = \Sigma_2$
$= \{(x_1, x_2, \ldots) : x_i \geq 0 \text{ and } \sum x_i = 1\}$

Fix $\sigma_1 \in \Sigma_1$ and fix $\epsilon > 0$, $\exists\, k \in \mathbb{N}$

$\sigma_1(\{1, \ldots, k\}) > 1 - \epsilon$

$u_1(\sigma_1, k+1) = \sigma_1(\{k+2, \ldots\}) 1 + \sigma_1(\{1, \ldots, k\})(-1)$

$< \epsilon(1) + (1-\epsilon)(-1) = -1 + 2\epsilon$

$\inf_{\sigma_2 \in \Sigma_2} u_1(\sigma_1, \sigma_2) \leq -1 \qquad \forall\, \sigma_1 \in \Sigma_1$

$u_1(s_1, s_2) \in \{-1, 0, 1\}$

$\inf_{\sigma_2 \in \Sigma_2} u_1(\sigma_1, \sigma_2) \geq -1 \qquad \forall\, \sigma_1 \in \Sigma_1$

$$\inf_{\sigma_2 \in \Sigma_2} U_1(\sigma_1, \sigma_2) = -1 \qquad \forall\ \sigma_1 \in \Sigma_1$$

$$\sup_{\sigma_1 \in \Sigma_1} \inf_{\sigma_2 \in \Sigma_2} U_1(\sigma_1, \sigma_2) = -1$$

$$\overline{U}^M = \inf_{\sigma_2 \in \Sigma_2} \sup_{\sigma_1 \in \Sigma_1} U_1(\sigma_1, \sigma_2) = 1$$

So, the value does not exist.

For mixed strategies — Two player zero sum game

$$\underline{U} = \sup_{\sigma_1 \in \Sigma_1} \inf_{\sigma_2 \in \Sigma_2} U(\sigma_1, \sigma_2)$$

$$= \sup_{\sigma_1 \in \Sigma_1} \left[\min_{s_2 \in S_2} U(\sigma_1, s_2) \right]$$

$$= \max_{\sigma_1 \in \Sigma_1} \min_{s_2 \in S_2} U(\sigma_1, s_2)$$

$$\overline{U} = \min_{\sigma_2 \in \Sigma_2} \max_{s_1 \in S_1} U(s_1, \sigma_2).$$

$|S_2| = m_2$

$$\Sigma_2 = \left\{ (u_1 \dots u_{m_2}) \;\middle|\; u_i \geq 0,\ \sum_{i=1}^{m_2} u_i = 1 \right\}$$